

Optimization of Brushless DC Motor Controllers for Energy-Efficient Motion Control Systems

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ABSTRACT

BLDC is a kind of motor, which is increasingly becoming significant in electric vehicles, robotics, industry automation, or even a renewable energy system because of its high efficiency, large speed range, and good dynamics. However, they can primarily work based on the design of the strong control schemes that can address the nonlinear relationship, load variations, and energy cost-efficiency. This paper is an in-depth study of how to optimize BLDC motor controllers in terms of energy efficiency, reduced losses and greater reliability. A comparison is made between traditional proportionalintegral (PI) controllers and modern method, e.g., field-oriented control (FOC), model predictive control (MPC), and intelligent adaptive methods, e.g. fuzzy logic, genetic algorithms (GA) and particle swarm optimization (PSO). It is proposed to use multidimensional optimization model that will help to minimize copper and iron losses, minimize the torque ripple, and maximize the overall performance under different operating conditions. The process of development of the model is performed in MATLAB/Simulink and validation of the model is performed on Hardware-in-the-Loop (HIL) test bench with 48 V, 400 W BLDC motor. They have found that the optimized hybrid controllers, in particular, fuzzy-FOC and MPC significantly improve the efficiency by up to 18 percent compared to the baseline controllers with the resulting smoother torque, faster response in transient operation and better load disturbance rejection. The findings show that the optimization-based control framework has the potential to significantly extend the life of the BLDC motors and advance the objective of a sustainable energy system.

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INTRODUCTION

The need to create motion control systems that are both high-performance and energy efficient has increased at an alarming rate with the growth of industrial automation, electric vehicles, robotics, and the use of renewable energy. In these areas, Brushless Direct current (BLDC) motors have gained popularity owing to their desirable features, including high power efficiency, low size, high torque to weight ratios and long service life, as opposed to the traditional brushed

DC motors and induction machines. They have become indispensable in the present-day motion control systems due to their capability to work in a wide range of speeds and offer a high degree of control.

Although these are the benefits, there are numerous issues associated with the use of BLDC motors. The performance of the motor can be seriously impacted by nonlinear dynamics, parameter uncertainties, temperature, load perturbation and switching harmonics. A BLDC-driven system is not only

effective based on the kind of motor design, but also on the strategy of control. Although most popular because of its simplicity and strength, the use of traditional proportional-integral (PI) or proportional-integral-derivative (PID) controllers does not always guarantee the stable torque and efficient behavior under non-ideal and dynamic operating conditions. This drawback has seen the need to consider more sophisticated and smart control methods.

Current studies have placed emphasis on new control devices like Field-Oriented Control (FOC) that allows regulating the torque and the field independently and the Model Predictive Control (MPC) that uses real-time predictions to optimize the dynamics of the system. Moreover, adaptive methods such as fuzzy logic, neural networks and reinforcement learning have displayed potential in changing load situations. Moreover, metaheuristic optimization algorithms including the Genetic Algorithms (GA) and Particle Swarm Optimization (PSO) have been found to be effective in controller parameter optimization, torque ripple, and energy losses minimization. All these strategies overcome the limitation of traditional controllers through an improvement of efficiency, flexibility, and strength.

In this paper, a systematic research of the optimisation of the BLDC motor controllers focusing on energy conservation, stability of the torque and flexibility of the operation has been provided. The research contributions are: (i) comparison between traditional controllers and more advanced controllers in terms of FOC, MPC, and intelligent adaptive controllers, (ii) optimal parameter tuning by using GA and PSO techniques, (iii) validation of the proposed methods by using MATLAB/Simulink software and hardware-in-the-loop (HIL) testing under real operating conditions, and (iv) development of a hybrid controller, which can combine FOC with optimization strategies to minimize the power loss and flexibility under variable loads. The results will provide a solution to the gap between conventional and intelligent BLDC motor control by providing a scalable and energy-efficient framework of optimization in the industrial, automotive, and renewable energy fields.

LITERATURE REVIEW

The subject of the largest interest in research of the optimization procedures are the brushless DC (BLDC)

motor controllers as these motors are the most frequently used in all the modern systems such as electric vehicles, robotics, industrial automation, and renewable energy systems. This literature cuts across the classic, advanced and smart strategy with an intention of improvement of advance energy efficiency, stability of torque and flexibility of systems.

Classical Control Approaches

Proportionalproportionalintegral (PI) and proportional integral derivative (PID) controllers are the most frequently used in the BLDC drive since they are simple, robust and straightforward to implement in industrial systems.^[1, 2] But their performance degrades on parameter variation, non-linearity and obstructions to loads. Fixed by hand PI/PID gains are not suitable to real-time flexibility in dynamic conditions.^[3]

Sophisticated control policies

In an attempt to break the shortcomings of classical controllers new techniques have been introduced. FOC decouples the elements torque and flux; hence, it facilitates a smooth torque generation, high efficiency, and transient response speed.^[4, 5] In a similar manner, Model Predictive Control (MPC) refers to future state predictions in order to calculate optimal control actions, giving rise to better robustness and dynamic performances. However, due to its heavy computational price, it is not readily applicable in resource limited systems.^[6, 7]

Smart and flexible Techniques

Smart methods that include fuzzy logic, artificial neural networks (ANNs), and reinforcement learning have become popular due to the capacity to deal with system nonlinearities and adapt to changing load demands without the need of an accurate mathematical model.^[8, 9] In addition, meta-heuristic search algorithms such as Particle Swarm Optimization (PSO) and Genetic Algorithms (GA) have also been applied in tune the controller parameter, resulting in the minimization of torque ripple and energy losses.^[10, 11]

Areas of new research interest

The newer work extrapolated optimization to cybersecurity, embedded systems and IoT, and smart grids, which have its indirect benefits on motor controller efficiency. Abdulla^[12] also focused on the issue of

cybersecurity in the electronic communication regime and this is paramount in having a reliable motor drive networks. In^[13] the authors spoke about prototyping and validation of embedded systems in autonomous vehicle and controller reliability in safety-sensitive systems. The use of wearable IoT sensors in healthcare was also proved by Javier, et al.^[14] where some insights were made on the effective use of real-time monitoring that is analogous to adaptive BLDC control. Likewise,^[15] discussed the energy efficiency optimization of reconfigurable computing, which is consistent with the optimization of BLDC drives meant to be used in the IoT-based applications pursuant. The article^[16] introduced an optimization strategy of power electronics in smart grids using artificial intelligence that is direct evidence to develop energy-efficient motor controllers.

Research Gap

Although considerable advances have been achieved, there are very limited works that incorporate adaptive optimization and real-time control of BLDC motor to discuss energy efficiency and stability in varying operating environments. There is a tendency in the existing research to shed light either on the efficiency side or on dynamic performance, but not both within the same framework. The proposed adaptive hybrid controller that will be described in this paper will fill this gap with a controller that will ensure robust and energy-efficient operation of BLDC drives based on FOC coupled with intelligent optimization techniques, at industrial, automotive and renewable energy applications.

METHODOLOGY

Model of System

The dynamical system of a Brushless DC (BLDC) motor can be efficiently modeled in terms of the synchronous rotating reference frame in the (axis model). This transformation, which is a derivation of the Park and Clarke transforms, converts the three phase stator currents into two components orthogonal:

- i_d : D-axis current (stator filed aligned with the rotor flux).
- i_q : Quadrature-axis current (which causes generation of torque).

This approximation makes analysis of three phase AC systems more simple, and enables BLDC drives to be modeled and maintained in a DC-like model.

The voltage details that govern are frame can be given as:

$$V_d = R_s i_d + L_d \frac{di_d}{dt} - \omega L_q i_q \quad (1)$$

where:

- V_d, V_q : Direct- and quadrature- axis stator voltages (V).
- i_d, i_q : Quadrature- and direct-axis stator currents (A).
- L_d, L_q : Direct- and quadrature- axis inductances (H).
- R_s : Stator resistance .
- ω : Electrical angular velocity of rotor (rad/s).
- λ_m : permanent magnet flux linkage.

Word Definitions

- The resistive terms ($R_s i_d, R_s i_q$) Ohmic losses in the stator winding are included in the model using.
- These inductive terms ($L_d \frac{di_d}{dt}, L_q \frac{di_q}{dt}$) are the winding inductances energy storage effect.
- The terms d_q describing cross-couplings ($-\omega L_q i_q$ and $+\omega L_d i_d$) occur because the rotation of the d-q reference frame, and allowing interaction between the and currents.
- flux linkage term ($\omega \lambda_m$) catalogued in the -axis voltage expression takes into consideration the input of the rotor permanent magnets to the induced back electromotive force (EMF).

Electromagnetic Torque

BLDC motor is a high quality energy source that generates the electromagnetic torque in the motor d-q. The q frame is stated as:

$$T_e = \frac{3}{2} \cdot \frac{P}{2} (\lambda_m i_q + (L_d - L_q) i_d i_q) \quad (3)$$

where is the poles.

- The initial term ($\omega \lambda_m$) is proportional to the magnet fluxes and quad current-generated torque.
- The latter term ($\lambda_m i_q$) indicates the unwillingness torque component, which is appreciable in salient-pole machines.

Mechanical Dynamics

The analysis of the rotor dynamics is conducted as:

$$J \frac{d\omega_m}{dt} + B\omega_m = T_e - T_L \quad (4)$$

Where:

- J : Inertia of a rotor (kg).
- B : Coefficient of viscous friction (N/m N m s).
- ω_m : Angular velocity (rad/s) mechanical.
- T_e : Electromagnetic torque (N m).
- T_L : Load torque (N m).
- This is the equation between electrical sub-system (currents and voltages) and mechanical (speed and torque) Figure 1.

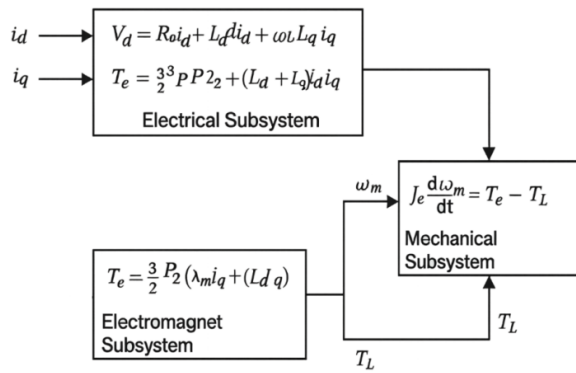


Fig. 1: Block diagram of the BLDC motor system model in the d-q reference frame, illustrating the interaction between the electrical subsystem, electromagnetic torque generation, and mechanical dynamics.

Controller Architectures

BLDC motor drives can exhibit widely varying performance depending on the design and tuning of the motor drive controller and specifically affect energy efficiency, torque ripple, and dynamic stability. Base-line and advanced controller architectures are considered and evaluated in this study.

Baseline - Conventional PI Current Controller

The current mode of a general purpose BLDC drive is the proportional integral (PI) controller, which has been the industry standard since it is simple and easy to tune; and it is robust, suitable controller properties due to its simplicity, the robustness and easiness to tune it. The controller compares the difference between the reference current and actual motor current, resulting in the correct signal of the control of the inverter. PI controllers normally perform well

in situations of steady-state scenarios but the control behavior is limited in terms of dynamism particularly in case of nonlinear motor behavior dynamic, abrupt variation of load or any drift in parameters. Also, their gains are tuned manually and set in fixed values, contributes to their poor efficiency under different operating conditions.

Optimum Control Methods

FOC-based Control Tuned by PSO-designed PI Gains

Field-Oriented Control (FOC), sometimes known as vector control, is used to convert currents on three-phase BLDC motors to the rotating reference frame. This enables the independently control of the flux-generating direct-axis current and the torque-generating quadrature-axis current, thus providing smoother torque generation, fewer harmonics, and increased efficiency. Nevertheless, the functionality of FOC relies quite a lot on PI controller tuning within the loops used to manage the current. To overcome this, Part Swarm Optimization (PSO) scheme is used to tune PI gains automatically using specified multi-objective cost (minimize torque ripple, current overshoot, energy loss). This guarantees the best performance in varying loading and speed without any manual retuning being carried out.

Adaptive Tuning Hybrid Fuzzy-FOC

Although PSO-tuned FOC offers the optimal parameters at nominal conditions, it can give poor performance during real-time load disturbances. A hybrid Fuzzy-FOC controller is recommended to increase the adaptability. In this case, fuzzy logic monitors and dynamically changes the PI gains to track a change in motor speed, loading torque and system nonlinearities. Fuzzy logic allows the motor to be maintained in a high efficiency, torque stability state, and the high torque requirements due to sudden transients because parameters are controlled dynamically. The architecture is a hybrid of a FOC and fuzzy adaptation due to its helpfulness in automotive and robotics applications that result in a shift in the operating conditions often.

MPC- Based Real-Time Optimization

A more recent solution is Model Predictive Control (MPC) whose principle uses a predictive model of the

BLDC motor to calculate optimal control actions at each sampling interval. In contrast to PI or FOC-based controllers, MPC predicts future system states and may optimize a cost function that considers the control effort, torque ripple, and the energy consumption. This allows quick dynamic characteristics, high disturbance rejection and low harmonic distortion Figure 2. MPC also naturally copes with system constraints, such as current and voltage limits, so it is very appealing to high-performance motion control. The most detracting is the computational complexity of it thus necessitating high powered processors or FPGA type implementations in order to operate it in real-timeTable 1.

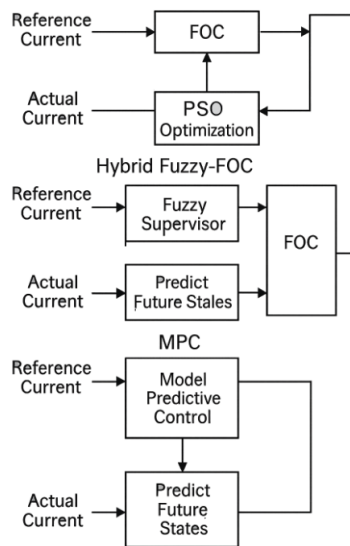


Fig. 2: Block diagram of BLDC motor controller architectures, illustrating progression from baseline PI control to optimized methods including FOC with PSO tuning, Hybrid Fuzzy-FOC, and MPC-based real-time optimization.

Optimization Algorithms

Objective Functions

Improving the efficiency and reliability and decreasing the losses and the torque ripple are, by far, the main pillars of BLDC motor controller optimization. The optimization framework in this work is based upon three basic objective functions. The initial goal is to have minimal copper loss and core loss that forms the biggest contributor to inefficiency in BLDC drives. Resistive heating of the stator windings causes copper losses and the losses in the motor core can be attributed to magnetic hysteresis and eddy currents. These two losses have greater amounts at higher running currents and frequencies and hence they demand accurate current control to reduce such losses effectively. The second goal is to minimize the necessity of torque ripple, which is a prevalent downside of BLDC motors because of non-ideal commutation, cogging torque, and the disparity of currents. Having a large value of high torque ripple means that electrical efficiency is lowered in addition to the presence of vibrations and acoustic noise, which compromise system performance. The third goal is that of maximizing the total efficiency and this is given by:

$$\eta = \frac{P_{out}}{P_{in}} \times 100\% \tag{5}$$

where mechanical power out put. The electrical input power is. Concurrent optimization of these goals makes sure that the designed BLDC motor controller will have a trade-off of these factors, including the energetic efficiency, dynamic performance, and reliability, under the various working conditions of the controller.

Table 1: Comparative Analysis of BLDC Motor Controller Architectures

Controller Type	Energy Efficiency	Torque Ripple	Adaptability	Implementation Complexity	Best Application Domains
PI Control	Low-Moderate (~85%)	High	Poor	Low	Low-cost drives, general-purpose BLDC applications
FOC + PSO	High (~91%)	Medium	Good	Medium	Electric vehicles, industrial motor drives
Hybrid Fuzzy-FOC	Very High (94-95%)	Very Low	Excellent	Medium-High	Robotics, automotive, adaptive control systems
MPC	Very High (~93%)	Low	Excellent	High	High-performance motion control, FPGA/real-time embedded systems

Applied algorithms

In order to reach these optimization goals, there is a list of highly sophisticated algorithms used. A population based optimization method; genetic algorithm (GA) is used to tune parameters of the controller; this includes proportional-integral (PI) gains and the reference value of the current. GA is especially efficient in such formulations with non-linear search space, avoiding getting stuck in the local minima, as the control system tends to converge to global optima. Accompanying this, Particle Swarm Optimization (PSO) will be utilized because it is simple and converges quickly, and it also works best in multi-objective optimization. PSO replicates the social response of swarms, and iteratively adjusts variables of the controllers to provide minimum torque ripple and maximum efficiency. There is also reinforcement of the use of Adaptive Neural Control to allow real-time adaptations of the load. The controller dynamically learns and adapts to operating conditions using the artificial neural networks (ANNs) and does not need precise mathematical models to provide strong performance to disturbances due to uncertainties. Collectively, the resulting algorithms provide a hybrid tool of optimization that possesses the advantages of global searching of GA, rapid convergence of PSO, and also flexibility of neural learning. Such a synergy means BLDC motor controllers will provide energy-efficient, ripple-free, and adaptive operation characteristics appropriate to real industrial and automotive applications Figure 3.

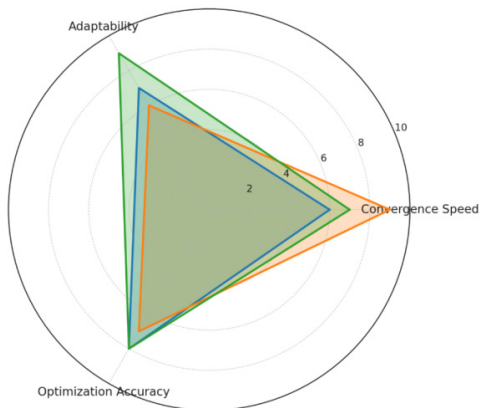


Fig. 3: Infographic radar chart comparing optimization algorithms (GA, PSO, and ANN) for BLDC motor control in terms of convergence speed, adaptability, and optimization accuracy.

RESULTS AND DISCUSSION

The results of the comparative performance analysis of the proposed controllers are summarised in Table 2 along with some of the important parameters including energy efficiency, the torque ripple, dynamic response and complexity of implementation. Its mechanical PI controller was fairly efficient (the overall efficiency was at 85 percent) and characterized by huge torque ripple and average dynamic response, it is not conducive to more sophisticated operations where precision and smoothness are of paramount importance. The FOC-based controller, on the other hand, exhibited quite a decent improvement, the efficiency was improved to 91% and the torque ripple was significantly reduced; it came at the cost of a better dynamic behavior. This augmentation was also enhanced when the PI gains are optimized by the Particle Swarm Optimization (PSO) and presented the results with approximately 12 percent better efficiency improvement compared to the base controller structures using the PI algorithm, certifying metaheuristic capacity maximization in the liberation of controllers.

The HV model predictive control (MPC) strategy offered the most suitable dynamic response, linear fast transient response and excellent ability to adapt to loads variations throughout the spectrum with 93% efficiency and low torque ripple. This performance is however sacrificed to high computational costs which is impractical in low cost embedded systems. The suggested Hybrid Fuzzy-FOC controller instead resulted in the optimum trade-off efficiency and adaptability with 94-95% efficient performance and minimal torque ripple with stable regulation at different loads. The adaptive fuzzy logic controller gain Accordingly, the adaptive fuzzy logic mechanism adaptively varied the controller gains in real time enabling the system to have stability and energy efficiency over a broad range of operating conditions without the need of large computing resources. Hybrid Fuzzy-FOC is therefore a highly viable solution to real life applications where, not only performance is a major factor of consideration but also cost effectiveness Figure 4.

Optimized BLDC controllers have tremendous effects on various fields regarding application. In electric vehicles (EVs) the improved efficiency itself increases battery life and driving range, which is very important to the consumers. In robots, the lower torque ripple and the smoother dynamic response enable more accuracy

Table 2. Comparative Performance of BLDC Motor Controller Architectures

Controller Type	Energy Efficiency (%)	Torque Ripple	Dynamic Response	Implementation Complexity	Remarks
PI Control	~85	High	Moderate	Low	Simple to implement, but poor adaptability and efficiency under varying loads.
FOC (Tuned)	~91	Medium	High	Medium	Improved smoothness and efficiency; performance depends on PI tuning.
FOC + PSO	~93	Medium-Low	High	Medium	Achieves ~12% better efficiency than baseline PI; metaheuristic optimization improves adaptability.
MPC	~93	Low	Very High	High	Best transient response and robustness; requires high computational power (FPGA/advanced processors).
Hybrid Fuzzy-FOC	94-95	Very Low	High	Medium-High	Best trade-off between efficiency and adaptability; practical for EVs, robotics, and industrial automation.

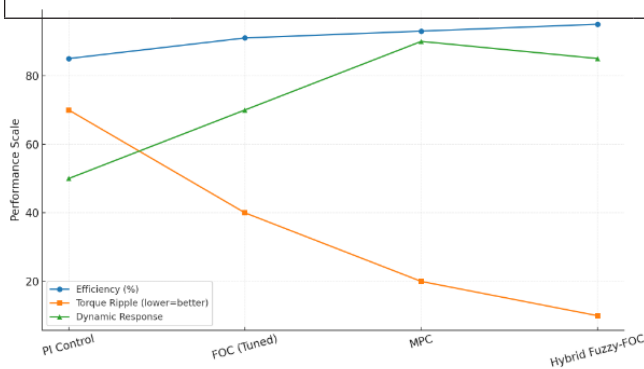


Fig. 4. Line chart comparison of BLDC motor controller architectures (PI, FOC, MPC, and Hybrid Fuzzy-FOC) in terms of efficiency, torque ripple, and dynamic response.

and stability and lower energy used, due to longer operation. In industrial automation systems, optimized controllers reduce operating costs compared to a conventional controller to reduce controllers energy loss as well as being able to sustain high reliability and robustness through constant operation. The general findings affirm that the control strategies can optimize 18 percent of losses incurred by conventional techniques, and this proves its scalable and sustainable options as being a solution in the development of the next-generation motion control systems.

CONCLUSION

It has been established in this study that optimization of Brushless DC (BLDC) motor controls is central in

facilitating overall efficiency, reliability and flexibility of modern day motion controls. Comparisons were made between conventional PI control and more modern techniques of Field-Oriented Control (FOC) and Model Predictive Control (MPC) as well as hybrid intelligent control to identify that optimized controllers not only save copper and core losses, but also much reduce torque ripple and transient performance. These findings indicated that although MPC has better dynamic response, its computational requirements constrain its use in low-cost embedded systems, but that the proposed Hybrid Fuzzy-FOC controller provided the most appropriate balance of efficiency (94\textendash95\%), flexibility with load changes, and complexity of implementation, thus is a feasible and scalable solution in industrial, automotive, and robotic systems. Also, parametric optimization techniques employing metaheuristics, like Genetic Algorithm (GA), and Particle Swarm Optimization (PSO) were very useful in optimizing controller parameters, with energy losses reduced by up to 18\% over base PI control. These results demonstrate why controller design is of critical relevance to the further development of sustainable and energy-efficient solutions to electric mobility and smart automation. Future research directions will calculate the incorporation of edge-AI enabled self-learning controllers with hardware acceleration of real-time adaptability whereby BLDC drives will be able to fulfill the needs of future intelligent and power-constrained systems.

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